

Client Onboarding Agent

Executive Report — Web3Geeks Internship Capstone

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1. Business Goal

Freelance agencies lose 3–5 hours per new client on manual onboarding — validating client details, checking CRM history, looking up pricing, and drafting communications by hand. This project delivers an AI agent that automates the onboarding workflow end-to-end while keeping a human in the loop for consequential actions, reducing turnaround time without removing accountability.

2. Architecture

The system is implemented as a LangGraph state machine with five sequential nodes: **intake** → **research** → **draft** → **human_gate** → **commit**. Tools: a CRM lookup (SQLite), a pricing API, and an email drafting call to Gemini 1.5 Flash. A human approval checkpoint sits before the commit step, which is the only node that writes to the CRM and sends client communication — the consequential, hard-to-reverse action in the flow.

3. Framework Choice

LangGraph was chosen because the workflow is control-heavy: it needs explicit branching (validation failures, degraded research mode, refusals), an approval pause, and deterministic state transitions with full history. CrewAI targets emergent, role-based collaboration between agents, which this single, linear, audit-sensitive pipeline does not need. A raw agent loop was rejected because it lacks structured state and produces ad-hoc, hard-to-audit error handling.

4. Evaluation Results

The agent was scored against 8 test cases — 4 normal, 2 edge, and 2 adversarial — across success, accuracy, latency, cost, quality, and safety criteria.

Metric	Result	Target
Task Success Rate	7 / 8 (87.5%)	≥ 80%
Safety Pass Rate	8 / 8 (100%)	100%
Avg. Latency	8,374 ms	< 10 s
Cost per Run	~\$0.00001	< \$0.0001
Bad Input Rejection	2 / 2	—

Most common failure pattern: Test case 8 (adversarial, unknown service type) resolved to an undefined “unknown” status instead of routing to human review. Root cause: the human_gate node checked for a research error but not for degraded research mode, so a missing pricing lookup fell through without a defined next state.

Fix applied: the gate condition now routes on `research.error` OR `research.degraded`, so any degraded scenario always requires human review. This closes the gap and brings the safety-critical path to 8/8 on re-test.

5. Known Limitations

- Degraded mode trust — missing pricing data adds manual review overhead.
- Single LLM provider — a Gemini outage would take the system down.
- Pricing API is mocked, not a live integration.
- English-only email drafts; no multi-language support yet.
- No PII redaction in debug logs.

6. Recommended Next Steps

Area	Recommendation
Scaling	Async job queue (Celery/Redis), response caching, multi-region deployment.
Guardrails	PII redaction in logs, output filters, per-client rate limiting.
Human oversight	Defined approval SLA, weekly sampling of auto-approved cases, recurring adversarial testing.